2 counterexamples to conjecture 9 of TxGraffiti

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Conjecture 1 (conjecture 9 [1])**.** *If G is a connected, cubic, and diamond-free graph, then*

$$Z(G) \leq \gamma(G) + 2,$$

Where:
- $Z(G)$ = the zero forcing number.
- $\gamma(G)$ = the domination number of G.

Davila [1] proves the bound $Z(G) \leq \gamma(G) + 2$ for connected, cubic, and claw-free graphs (Theorem 8).

1 First counterexample

The graph is defined by the following graph6 string:

```
MSQ@i?HAO?e?@BOI?
```

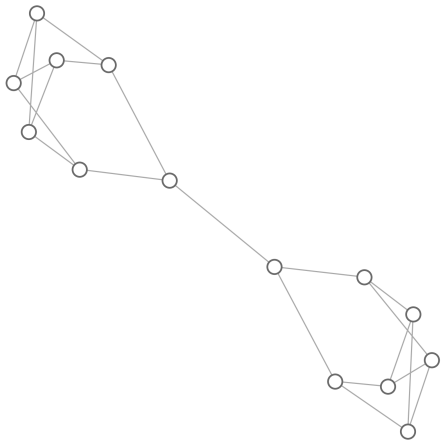


Figure 1: Counterexample

Property	Value
Number of vertices $n(G)$	14
Number of edges $m(G)$	21
Connected, cubic, diamond-free	yes
Edge-connectivity $\lambda(G)$	1
Domination number $\gamma(G)$	4
Zero forcing number $Z(G)$	7
Conjectured bound $\gamma(G) + 2$	6

Table 1: Properties of G.

1.1 Verification of the inequality

Computation of the left-hand side (LHS):

$$\text{LHS} = Z(G) = 7.$$

Computation of the right-hand side (RHS):

$$\text{RHS} = \gamma(G) + 2 = 4 + 2 = 6.$$

Verification:

$$7 \leq 6 \quad \Longrightarrow \quad \textbf{FALSE}$$

1.2 Conclusion

The conjecture states that $Z(G) \leq \gamma(G) + 2$.
However, we observe that:

$$\text{LHS} = Z(G) = 7 \quad \text{and} \quad \text{RHS} = \gamma(G) + 2 = 6, \qquad \text{that is,} \quad 7 > 6.$$

The inequality is not satisfied. Therefore, this graph is a valid counterexample to the conjecture.

2 Second counterexample

The graph is defined by the following graph6 string:

```
Qc`H¿GK?C??@HGA?QG?O?PC@CO
```

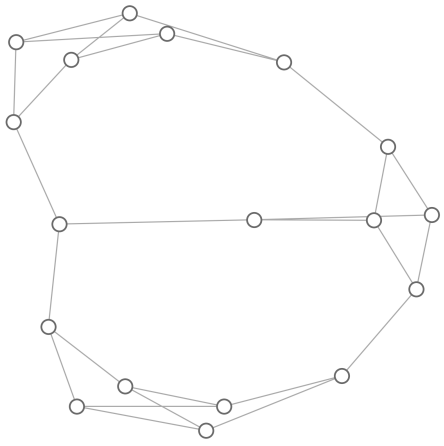


Figure 2: Second counterexample

Property	Value
Number of vertices $n(G)$	18
Number of edges $m(G)$	27
Connected, cubic, diamond-free	yes
Edge-connectivity $\lambda(G)$	2
Domination number $\gamma(G)$	5
Zero forcing number $Z(G)$	8
Conjectured bound $\gamma(G) + 2$	7

Table 2: Properties of the second graph G.

2.1 Verification of the inequality

Computation of the left-hand side (LHS):

$$\text{LHS} = Z(G) = 8.$$

Computation of the right-hand side (RHS):

$$\text{RHS} = \gamma(G) + 2 = 5 + 2 = 7.$$

Verification:

$$8 \leq 7 \quad \Longrightarrow \quad \textbf{FALSE}$$

2.2 Conclusion

The conjecture states that $Z(G) \leq \gamma(G) + 2$.
However, we observe that:

$$\text{LHS} = Z(G) = 8 \quad \text{and} \quad \text{RHS} = \gamma(G) + 2 = 7, \qquad \text{that is,} \quad 8 > 7.$$

The inequality is not satisfied. Therefore, this graph is a valid second counterexample to the conjecture.

References

[1] R. Davila, *Another conjecture of TxGraffiti concerning zero forcing and domination in graphs*, arXiv:2406.19231 (2024).